المعهد الهندي للتكنولوجيا دلهي
أبوظبي

INDIAN INSTITUTE OF TECHNOLOGY DELHI
ABU DHABI

ASBL100 Semester-I 2026

Introductory Biology For Engineers 3-0-2

Prof. Saurabh Raj
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Course Outline

- Darwinian evolution & molecular perspective
- Introduction to phylogeny
- Cellular assemblies – From single cell to multi-cellular organisms
- Comparing natural vs. humanmade machines
- Infection, disease and evolution
- Immunology
- Cancer Biology
- Engineering designs inspired by biology



Textbooks

- Molecular Biology of the Cell, Garland Science, Taylor and Francis Group, New York, Author: B. Alberts
- Lehninger principles of biochemistry, W.H. Freeman, Authors: Nelson, D. L., & Cox, M. M.
- Biochemistry, W.H. Freeman, Authors: Lubert Stryer, Jeremy Berg, John Tymoczko, Gregory Gatto
- Molecular Cell Biology, W.H. Freeman, Authors: Lodish, Harvey F.
- Medical Microbiology, Churchill Livingstone, Authors: Greenwood D, Slack R, Barer M, Irving WL
- Kuby Immunology, W.H. Freeman, Authors: Judith A Owen, Jenni Punt and Sharon A. Stanford
- Molecular Cloning, Cold Spring Harbor Laboratory Press, J. F. Sambrook and D. W. Russell

Overall Marks Distribution

Mid Semester	End Semester	Lab (experiment + theory)
35	35	25+5

⚠ Attendance rule: Below 75% may lead to one grade down!

Motivation

- Understanding health, disease, current and future therapies
- Systems biology/synthetic biology
- Integrated technologies and engineering for broader benefit
- Social science and ethics